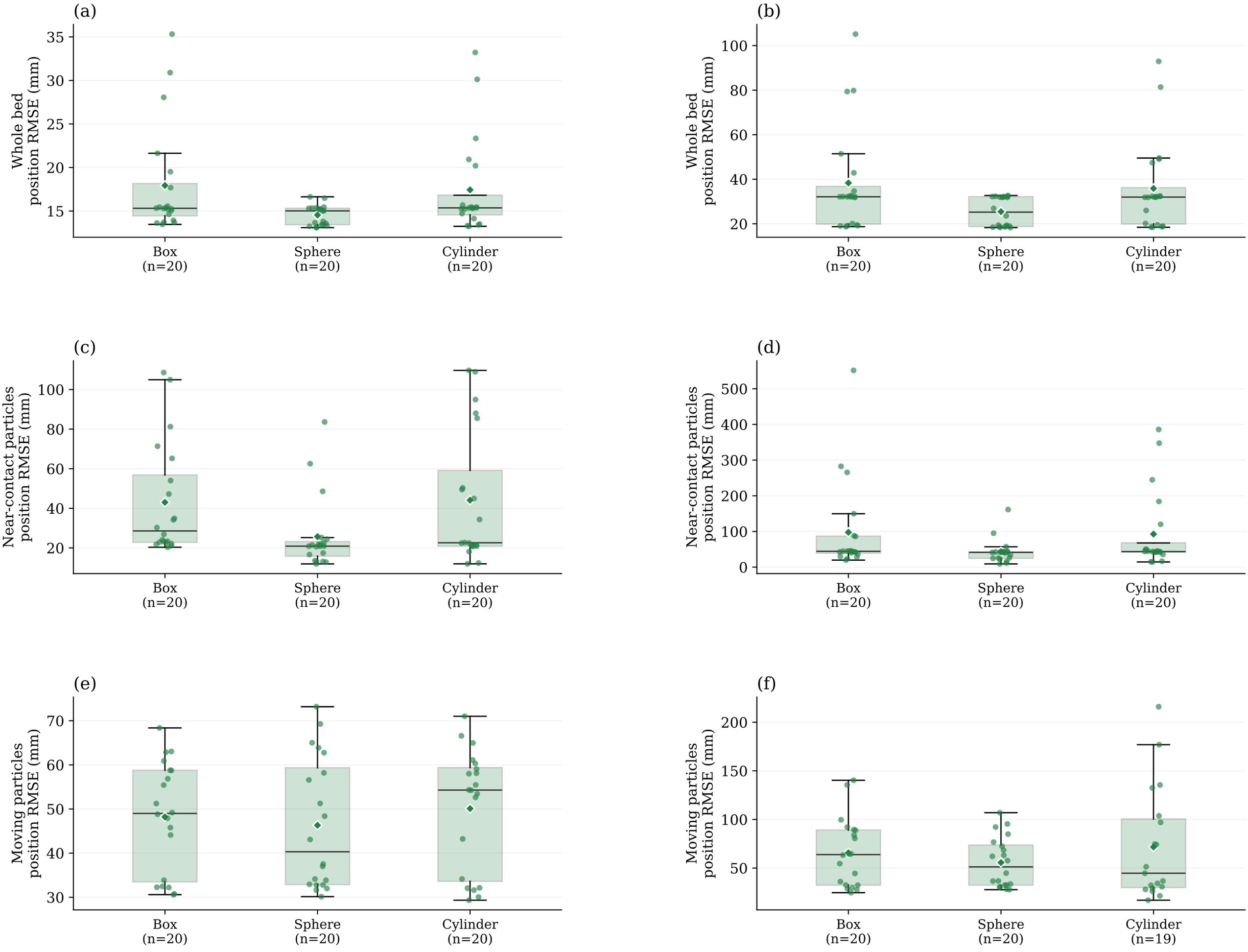

Figure 1: Frozen object-model checkpoint 9,000. (a,b) Whole-bed position RMSE; (c,d) near-contact particle RMSE; (e,f) moving-particle RMSE. Left: time mean over 0–6 s; right: final 6 s frame. NeRD errors use corresponding MPM particles. Near-contact reference distance ≤0.05 m; moving reference speed >0.01 m/s. One final moving region is empty and remains unavailable. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.